

introduction

- Air is essential for life it self, without it we could survive only a few minutes.
- It constitutes immediate physical environment of living organisms
- The atmosphere is layered in to four distinct which are: Troposphere, stratosphere, mesosphere, and thermosphere.

Normal components of our atmosphere

- ❑ **Nitrogen - 78.1%**
- ❑ **Oxygen - 20.9%**
- ❑ **Carbon dioxide - 0.03%**
- ❑ **Everything else - 0.07%**
 - noble gases (krypton, xenon, argon, helium)
 - methane
 - sulfur dioxide

What is air Pollution?

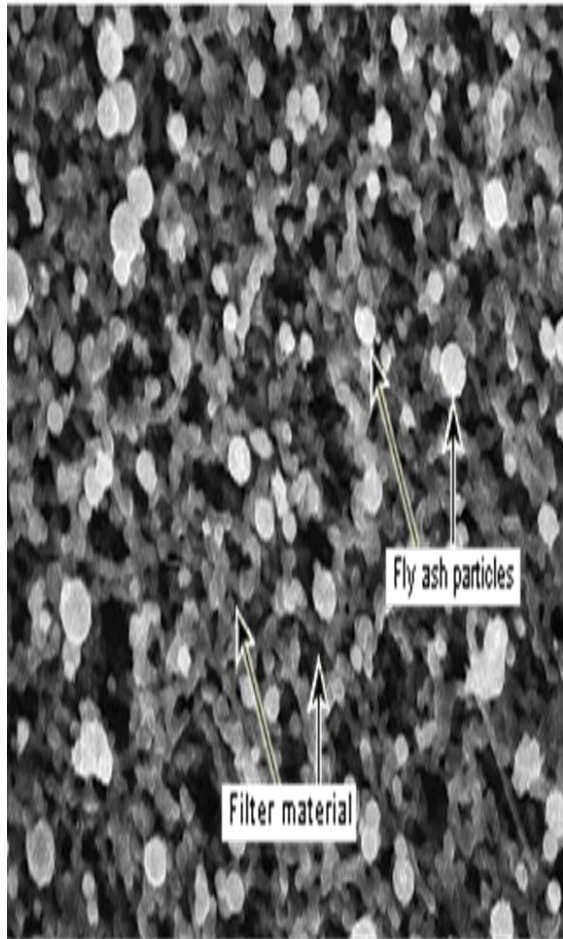
Air pollution consists of gases, liquids, or solids present in the atmosphere in high enough levels to harm humans, other organisms, or materials



Cont....

- may be defined as any atmospheric condition in which certain substances are present in such concentrations that they can produce undesirable effects on man and his environment.

Figure 1. Photomicrograph of Fly Ash Particles on an MCE Filter



Source: Courtesy of Air Control Techniques, P.C.

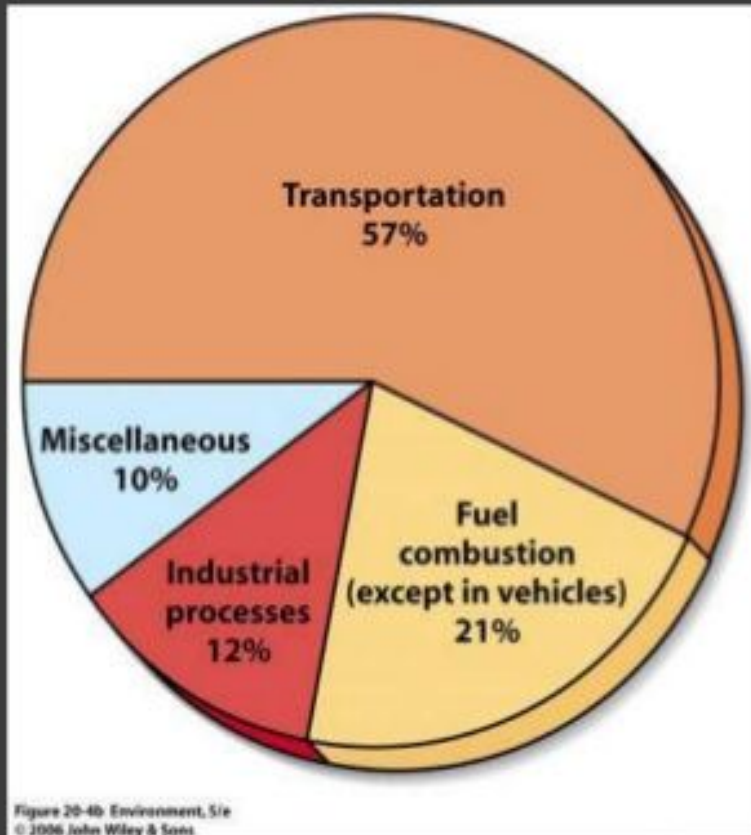
Particulate matter (PM)

describes a wide variety of airborne material. PM pollution consists of materials (including dust, smoke, and soot), that are directly emitted into the air or result from the transformation of gaseous pollutants. Particles come from natural sources (e.g., volcanic eruptions) and human activities such as burning fossil fuels, incinerating wastes, and smelting metals.

What is air Pollution?

- Air pollutants may be either emitted into the atmosphere or formed within atmosphere itself:
- Primary air pollutants:
Sources such as factory chimneys or exhaust pipe.
- Secondary air pollutants:
Ex ozone.

Sources of Primary Air Pollutants



The main sources of air pollution are:

(a) AUTOMOBILES:

- Motor vehicles are a **major source** of air pollution throughout the urban areas.
- They emit hydrocarbons, carbon monoxide, lead, nitrogen oxides and particulate matter.
- In strong sunlight, certain of these hydrocarbons and oxides of nitrogen may be converted in the atmosphere into "photochemical" pollutants of oxidizing nature.

The main sources of air pollution are:

(b) INDUSTRIES:

- Combustion of fuel to generate heat and power produces smoke, sulphur dioxide, nitrogen oxides and fly ash.
- Petrochemical industries generate hydrogen fluoride, hydrochloric acid and organic halides.

The main sources of air pollution are:

(c) DOMESTIC SOURCES :

- Domestic combustion of coal, wood or oil is a major source of smoke, dust, sulphur dioxide and nitrogen oxides.

(d) tobacco smoke:

- The most direct and important source of air pollution affecting the health of many people .
- Even those who do not smoke may inhale the smoke produced by others ("passive smoking").

The main sources of air pollution are:

(e) MISCELLANEOUS:

- These comprise burning refuse, incinerators, pesticide spraying, natural sources (e.g., wind borne dust, fungi, molds, bacteria) and nuclear energy programmes.

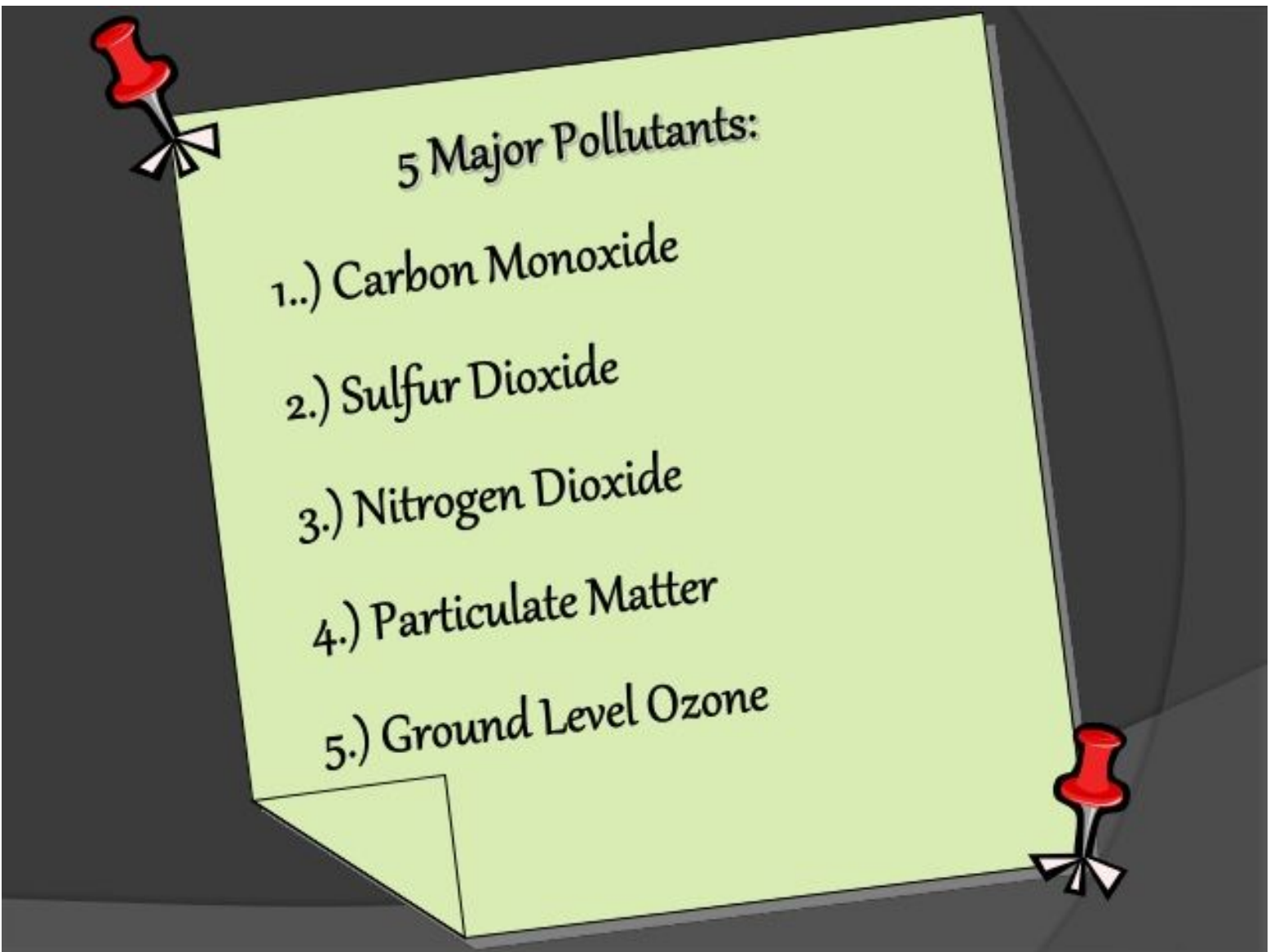
All these contribute to air pollution.

NOTE...

- Although the Earth's atmosphere extends to several layers above the surface, it is only the **first 30 km** that hold the major portion of the atmospheric gases.
- Man is most directly concerned with only the **8-10 km** of the atmosphere.
- The level of atmospheric pollution at anyone time depends **upon meteorological factors, e.g., topography, air movement and climate.**
- Winds help in the dispersal and dilution of pollutants.
- If the topography is dominated by mountains (or tall buildings) the winds become weak and calm, and pollutants tend to concentrate in the breathing zone.

NOTE...

- The vertical diffusion of pollutants depends upon the temperature gradient.
- When there is a rapid cooling of lower layers of air (temperature inversion), there is little vertical motion and the pollutants and water vapours remain trapped at the lower levels and the result is "smog".
- The "temperature inversion" which is more frequent in the winter months than in spring or summer, is a threat to human health.



5 Major Pollutants:

- 1.) Carbon Monoxide
- 2.) Sulfur Dioxide
- 3.) Nitrogen Dioxide
- 4.) Particulate Matter
- 5.) Ground Level Ozone

(1) Carbon monoxide

- It is colorless , odorless gas , a product of incomplete combustion of carbon containing materials, such as in in automobiles, industrial process, heating facilities and incinerators
- Some widespread natural non biological and biological sources have also been identified.
- Concentrations in urban areas depend on weather and traffic density.

Effect of CO pollution on the health:

- It causes harmful effect by reducing oxygen delivery to body organ , in externally high level it can cause death.
- CO's affinity for Hb is 240-270 times greater than oxygen and Fetal Hb has higher affinity for CO , so it competes with O₂ to bind (irreversibly) with haemoglobin.

Effect of CO pollution on the health:

- By this exposure to it reduce the oxygen-carrying capacity of the blood to the heart, brain and other organs .
- deprives body of O₂ causing headaches, fatigue, MI and impaired vision.

(2) Sulphur dioxide

- It is one of the several forms in which sulphur exists in air.
- The others include H_2S , H_2SO_4 and sulphate salts.
- Sulphur dioxide results from the combustion of sulphur containing fossil fuel, and when coal and fuel oil are burned.
- Domestic fires can also produce emissions containing sulphur dioxide.
- Acid aerosol - sulphuric acid (H_2SO_4) is a strong acid that is formed from the reaction of sulphur trioxide gas (SO_3) with water.

(3) Nitrogen Dioxide

- Nitric oxide (NO) is produced by combustion.
- Nitrogen dioxide (NO_2), which has greater health effects, is a **secondary pollutant created by the oxidation of NO under conditions of sunlight**, or may be formed directly by higher temperature
- combustion in power plants or indoors from gas stoves.

(4) Particulate matter :

- represents a complex mixture of organic and inorganic substances.
- Mass and composition tend to divide into two principal groups :
- coarse particles larger than $2.5\text{ }\mu\text{m}$ in aerodynamic diameter, and fine particles smaller than $2.5\text{ }\mu\text{m}$ in aerodynamic diameter.
- The smaller particles contain the secondarily formed aerosols ,combustion particles and recompensed organic and metal vapours.
- The large particles usually contain earth's crustal material and fugitive dust from roads and industries.

(5) Ground level ozone

- This is formed when pollutants such as nitrogen oxides and volatile organic compounds (VOCs) react in sunlight .
- High levels can cause breathing problems, reduce lung function and trigger asthma symptoms.
- Ground level ozone can also seriously damage crops and vegetation.
- Ozone is a powerful greenhouse gas and contributes to global warming both directly and by reducing carbon uptake by vegetation.

SMOG

- Smog is made up of mixture of those pollutants in the atmosphere .
- Combination of words smoke and fog.
- There are two types : reducing smog characterized by sulphur dioxide and particulate.
- And photochemical smog characterized by ozone and other oxidants.

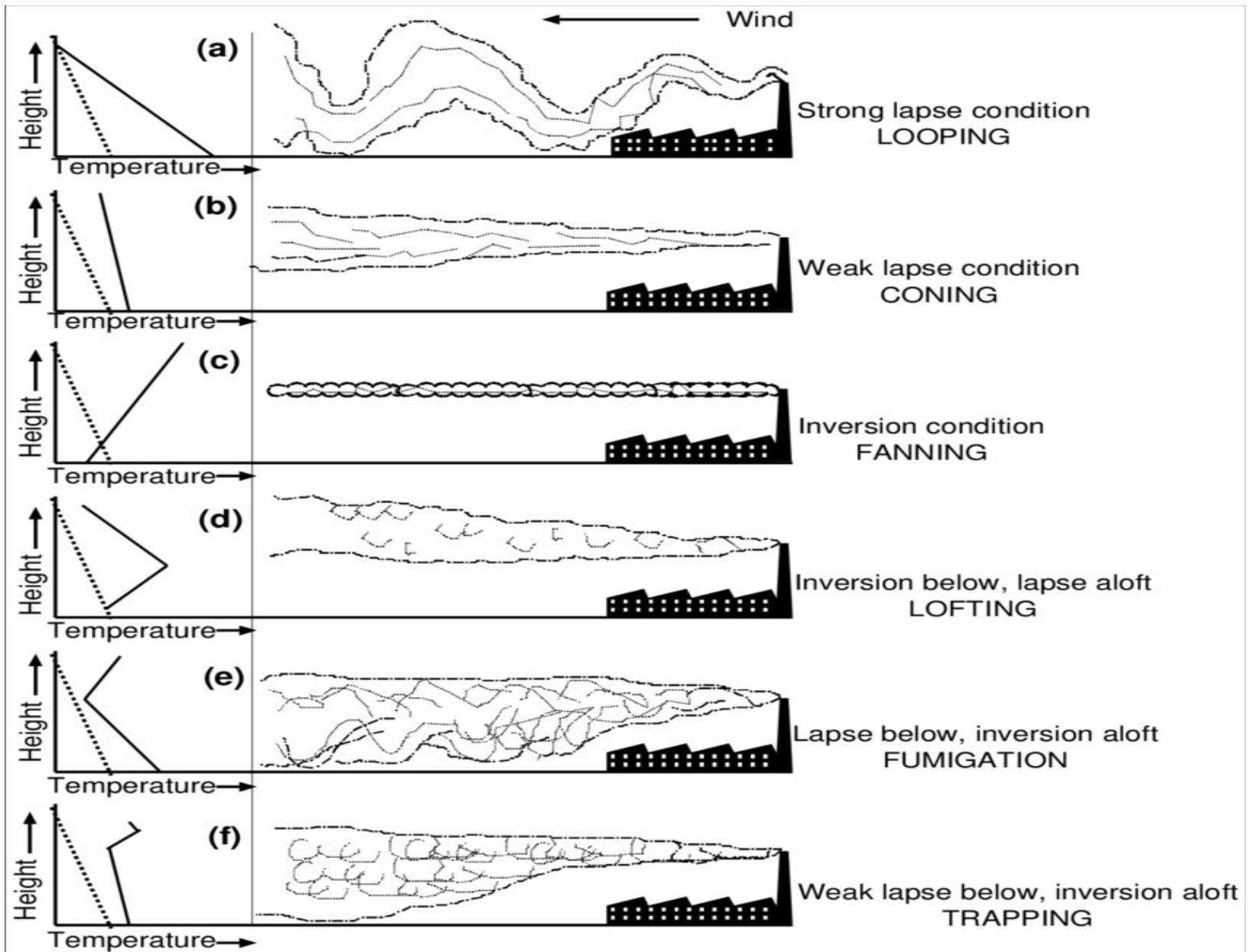


Carbon dioxide:

- ⦿ This is not commonly regarded as an air pollutant, although man generates enormous amount of it in combustion process using coal, oil and gas.
- ⦿ Carbon dioxide is a natural constituent of the air.
- ⦿ It does not take part in any significant chemical reactions with other substances in the air.
- ⦿ However, its global concentration is rising above the natural level by an amount that could increase global temperature enough to affect climate markedly

Volatile organic compounds:

- ⦿ VOCs are divided into the separate categories of methane (CH_4) and non-methane (NMVOCs)
- ⦿ Methane is an extremely efficient greenhouse gas which contributes to enhanced global warming
- ⦿ NMVOCs, the aromatic compounds benzene, toluene and xylene are suspected carcinogens and may lead to leukaemia through prolonged exposure .



Control of air pollution

CONTROL AT SOURCE

- 1. Source relocation**
- 2. Source shut down**
- 3. Fuel or Energy substitution**
- 4. Process changes**
- 5. Good operating practices**
- 6. Vehicular emission control**

PARTICULATE MATTER POLLUTANT

Particulate matter is the sum of all solid and liquid particles suspended in air, many of which are hazardous. This complex mixture contains for instance dust, pollen, soot, smoke, and liquid droplets.

Temperature and chemical composition of gases to be cleaned from particulants before the selection of proper control device.

Information on the physical form of the suspended material, its abrasive properties, size and shape, chemical composition and electrical sensitivity is required for this selection.

There Five major groups of processes.

- a) Settling chambers
- b) Inertial separator or Cyclone
- c) Electrostatic precipitator
- d) Bag houses and filters
- e) Wet scrubbers

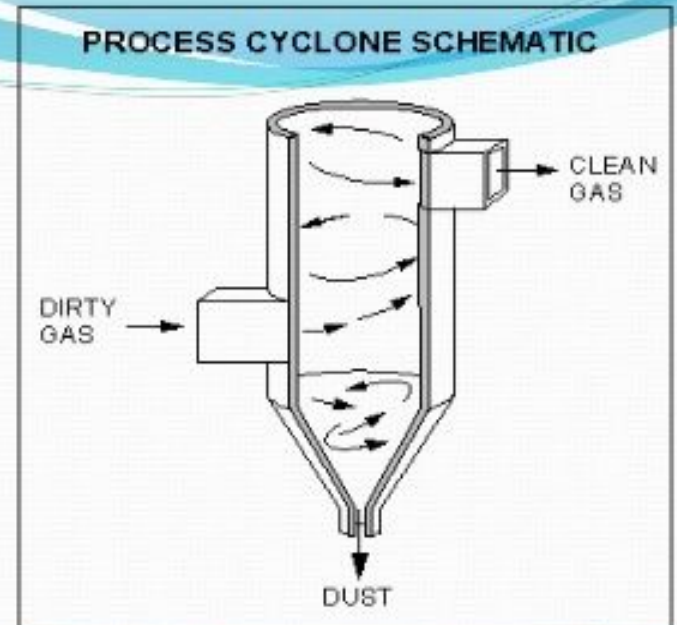
a) **SETTLING CHAMBERS**

- A simplest device, collecting dust of size $>10\mu\text{m}$.
- Settling chambers use the force of gravity to remove solid particles. The gas stream enters a chamber where the velocity of the gas is reduced. Large particles drop out of the gas and are recollected in hoppers. Because settling chambers are effective in removing only larger particles, they are used in conjunction with a more efficient control device.
- The size, shape of particles and density and viscosity are important parameters.



b) INERTIAL SEPERATOR OR CYCLONE

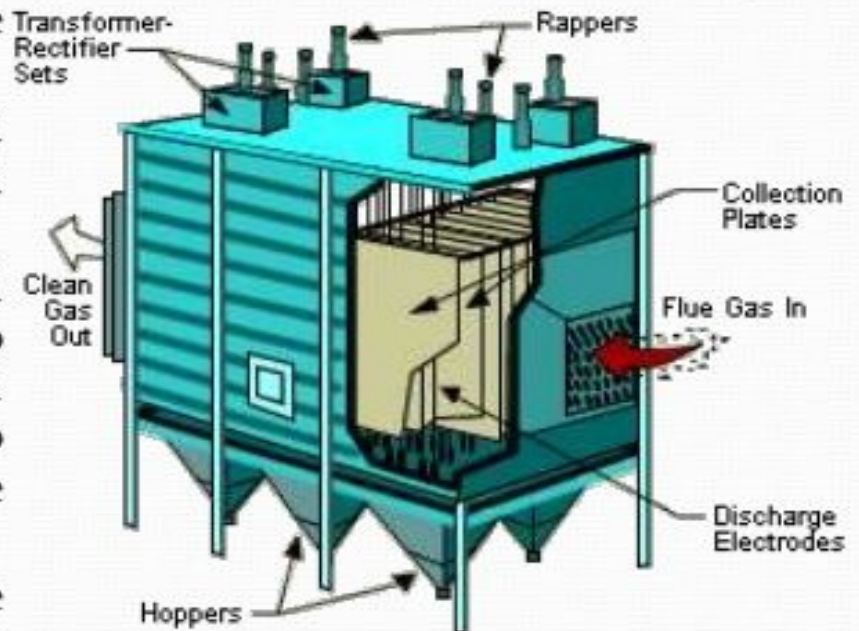
This process is achieved by a sudden change in the direction of gas flow. The dust particle is spinning in a circular path, it is subjected to an outward force. As a result the particles collect on the wall and fall to the bottom of the vessel.



c) ELECTROSTATIC PRECIPITATORS

- Passing the particle laden gases between high voltage discharge electrodes.
- Majority of particles get charged and collected on ground electrodes.
- At intervals both discharge and collecting electrodes are rapped to dislodge collected particles from the electrodes, which then fall into hoppers at the bottom of the precipitator.
- Ele. Precipitator is efficient for the particle size below $0.01\mu\text{m}$, can tolerate operating temperature as high as 700°K .

Figure 9. Conventional Electrostatic Precipitator



d) BAG HOUSE AND FILTERS

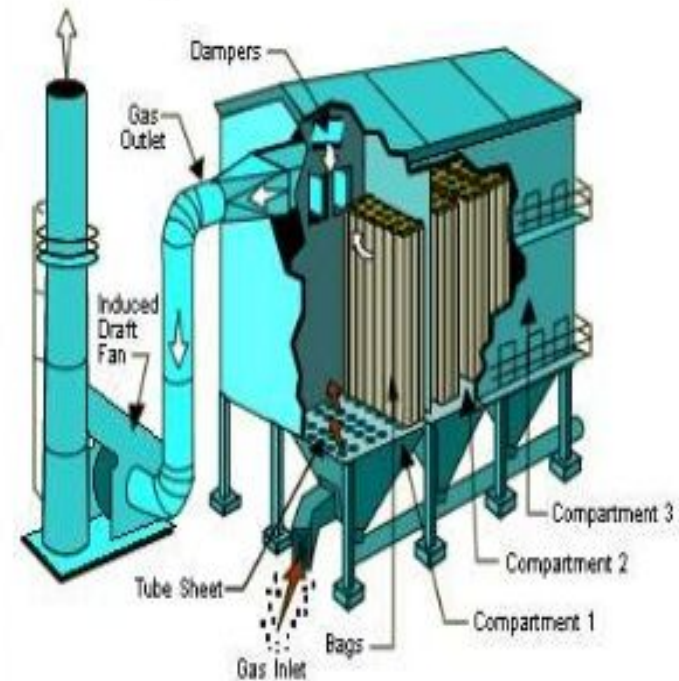
This method is most often used type of device, which is required to meet any future standards limiting the emission of respirable particles.

The collection efficiency is very high and even for particle of $0.01\mu\text{m}$ diameter.

The Fabric filters, or bag houses , remove dust from a gas stream by passing the stream through a porous fabric.

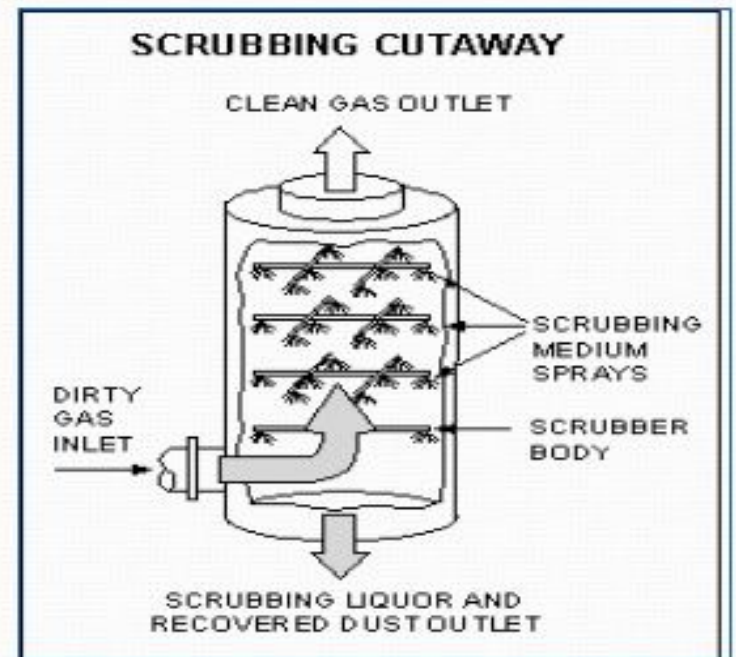
The fabric filter is efficient at removing fine particles and can exceed efficiencies of 99% in most applications.

Figure 13. Reverse Air Fabric Filter



E) WET SCRUBBERS

Objective of scrubber is to entrain the particulate matter in liquid droplets. Water subsequently flows from the bottom of the scrubber, the particulate is allowed to settle and clarified water is re-circulated.



- Scrubbers are used to remove acid gases, such as HCl and SO₂, and alkaline gases from the flue gas. There are several scrubber types: wet, semi-dry, and dry.
- The differentiating factor between these scrubbers is the environment in which the flue gas is treated.
- Wet scrubbers dissolve the flue gas in an aqueous washing solution, such as [sodium hydroxide](#) (NaOH), and remove HCl and HF in the first stage and SO₂ in the second stage at a pH close to neutral or alkaline.

- Semi-dry scrubbers spray the flue gas with an aqueous absorption agent as opposed to saturating it, and [dry scrubbers](#) pass the flue gas through an agent that is in the form of a fine, dry powder, such as lime.
- The acids react with the powder and the solid reaction products are filtered out. The advantage of wet scrubbers is that they are generally less expensive than semi-dry and dry scrubbers; however, the disadvantage is that they produce wastewater that must be treated and discharged.
- Semi-dry and dry scrubbers are less prone to corrosion and do not generate wastewater; however, they consume more chemicals and yield [solid residues](#)

2) GASEOUS POLLUTANTS

The removal methods for gaseous pollutants are designed to concentrate the pollutants in a liquid (absorption) or a solid (adsorption). Sometimes direct conversion of gaseous pollutants is possible by combustion.

The central pollution (prevention and control) board, New Delhi has fixed standard for ambient air quality in India under the Air Act, 1981 beyond which an ambient air can be considered polluted in a legal sense.

AMBIENT AIR QUALITY STANDARD IN INDIA **(concentration in μgm^3)**

Area category	Suspended particulate matter	SO ₂	CO	NO ₂
Industrial and mixed use	500	120	5000	120
Residential and Rural	200	80	2000	50
Sensitive area (tourist resort, game sanctuary)	100	30	1000	30

STANDARD FOR VEHICULAR EMISSION

Diesel driven vehicles: (Smoke density)

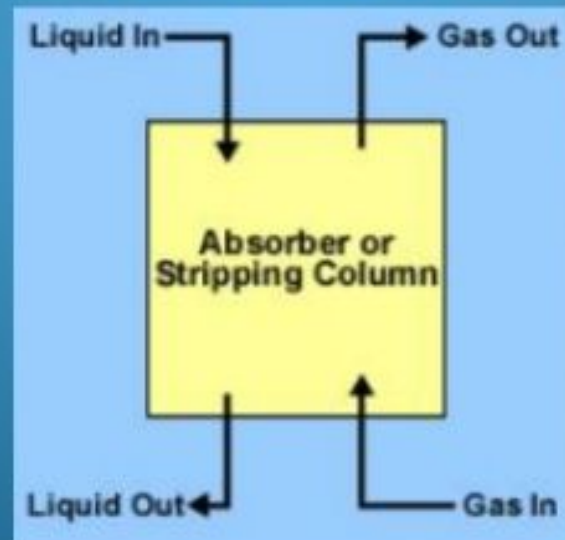
Vehicle at manufacturing stage	60 Hatridge Smoke Units (HSU)
Vehicle on road	70 HSU

Petrol driven vehicles: (Limits of CO in percentage volume)

Category	Vehicle on road	Vehicle at manufacturing stage
2 or 3 wheeler vehicles with engine displacement less than 50cc	5.0	5.0
Other 2 or 3 wheelers	4.5	5.5
4 wheelers	4.0	3.5

a) ABSORPTION

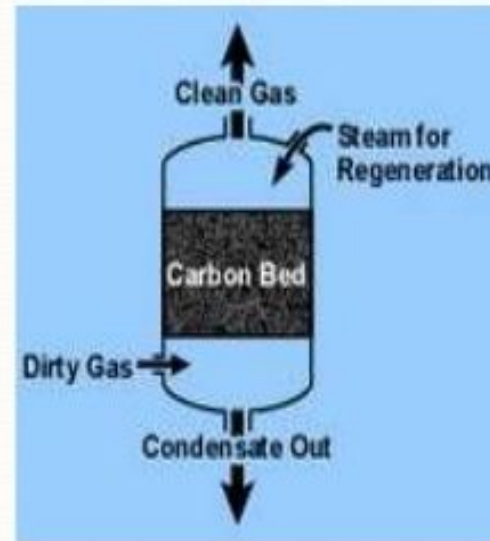
- This process is selected to remove gaseous pollutants by dissolution into a liquid solvent such as water or in a caustic or acid solution. (most commonly water is used as an absorbent).



b) ADSORPTION

Adsorption is a process where gases, vapours or liquids are concentrated on a solid surface as a result of surface or chemical force (physical adsorption and chemisorptions respectively). The amount of adsorbed substances depends directly on the internal surface area of solid and the kinetics of the process.

The most important adsorbents in industrial use to-days are Bauxite, activated carbon, activated alumina, silica gel and molecular sieves.



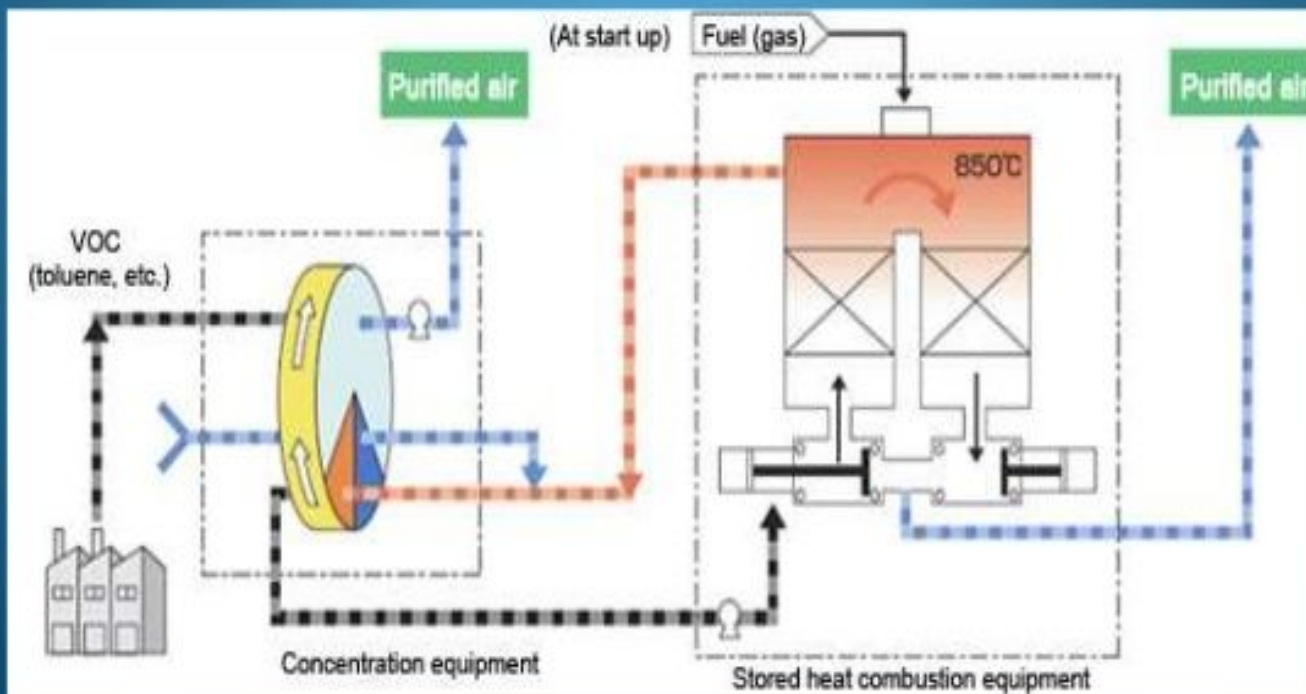
c) COMBUSTION

This method for the removal of VOCs since they can be decompose to CO_2 and H_2O .

The Direct and indirect or catalytic methods are used.

The catalytic method is preferred when low temperature of the process is desired.

The most commonly used catalysts are the Cu, Ni, V and Zn compounds. IF the concentrations of air pollutants are high, a combination of sorption methods and catalytic reaction is used.



- Catalytic reduction, [activated carbon](#), and dry sorbent injection are employed to reduce NO_x and remove heavy metals, dioxins/furans, and PM, respectively.
- NO_x can be destroyed via selective noncatalytic reduction (SNCR) or [selective catalytic reduction](#) (SCR). SNCR uses dry urea, NH_3 , as a reducing agent at temperatures between 900 and 1050°C to form water and N_2 while SCR uses a catalyst, such as a mixture of NH_3 and air, to form oxygen and water.

- SNCR is cheaper and yields less [corrosion problems](#) while SCR is more efficient, achieving reduction rates of up to 85%. Activated carbon and alkali sorbents such as lime and NaOH are injected into flue gas to remove volatile heavy metals and acid gases
- (IC) engines for motor vehicles are of two kinds: spark [ignition](#) (SI) and [compression ignition](#) (diesel). The SI engines use gasoline, alcohol, or compressed natural gas fluids (propane) or compressed natural gas as fuel; introduce the fuel to the cylinder by carburation or injection; and use a [spark plug](#) for ignition. Diesel engines use heavier fuels (usually designated diesel fuel), introduce the fuel to the cylinder by injection; and ignite the fuel–air mixture by the heat of compression of the cylinder.

- Spark ignition engines can operate either fuel-rich (more fuel than stoichiometrically necessary) or fuel-lean (more air than stoichiometrically necessary). Diesel engines are designed to operate fuel-lean without smoke emission and emit smoke when operated fuel-rich.
- The principal pollutants emitted by IC engines are carbon monoxide, oxides of nitrogen, unburned hydrocarbons and particulate matter (smoke

- Carbon monoxide and unburned [hydrocarbon](#) control is achieved by fuel-lean operation, improved [carburetor](#) calibration, [fuel injection](#), and catalytic conversion (afterburners). [Nitrogen oxide](#) control is achieved by control of [combustion chamber](#) temperature via [exhaust gas recirculation](#) and by catalytic conversion (afterburners). Additional unburned hydrocarbon control is achieved by control of fuel evaporation from the fuel tank and the carburetor and of cylinder blow-by by closure of the [crank case](#) and its venting by a positive crank case ventilation valve back to the carburetor.